

Physical Chemistry (I)

Lecture 12

Classical Statistical Mechanics



In the Last Class

- Maxwell-Boltzmann statistics
 - Boltzmann factor
 - Partition function
- Average quantities
 - Average energy
 - Average entropy: Shannon entropy
 - Partition function and thermodynamics

Last class

Maxwell-Boltzmann Statistics

Average distribution of independent particles
of a system in **thermal equilibrium**

Described by the **Boltzmann factor**

$$P(s) \propto e^{-E(s)/k_B T}$$

Partition Function

$$P(s) = \frac{1}{q} e^{-E(s)/k_B T}$$

Using the normalization condition, $\sum_s P(s) = 1$,

$$1 = \sum_s \frac{1}{q} e^{-E(s)/k_B T} = \frac{1}{q} \sum_s e^{-E(s)/k_B T}$$

$$q = \sum_s e^{-E(s)/k_B T}$$

(Partition function)

Last class

Average Energy

$$\langle E \rangle = \sum_i P(E_i) E_i = \sum_i \frac{E_i e^{-\beta E_i}}{q}$$

$$\frac{\partial \ln q}{\partial \beta} = \frac{1}{q} \frac{\partial q}{\partial \beta} = -\frac{1}{q} \sum_i E_i e^{-\beta E_i}$$

Hence,

$$\langle E \rangle = U = -\frac{\partial \ln q}{\partial \beta}$$

Last class

Average Entropy

$$S = -k_B \sum_i p_i \ln p_i \text{ (Shannon entropy)}$$

Substitution of $p_i = P(E_i) = \frac{1}{q} e^{-\beta E_i}$ leads to

$$S = \frac{\langle E \rangle}{T} + k_B \ln q$$

Therefore,

$$-k_B T \ln q = \langle E \rangle - TS = A$$

Partition Function and ...

- Helmholtz energy: $A = -k_B T \ln q$
- Internal energy: $U = -\partial(\ln q)/\partial\beta$
- Entropy: $S = (U - A)/T$

“A partition function contains
all the **thermodynamic information** about a system.”

Last class

Derived Functions

We know Helmholtz energy from the partition function,

$$A = -k_{\text{B}}T \ln q$$

Using $dA = -SdT - pdV$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_V, \quad p = -\left(\frac{\partial A}{\partial V}\right)_T$$

Example: Spin System

Assume that we have a single spin system: up ($\uparrow$) or down ($\downarrow$)

$$E(\uparrow) = +\frac{\epsilon}{2}, \quad E(\downarrow) = -\frac{\epsilon}{2}$$

The partition function is

$$q = \sum_s e^{-E(s)/k_B T} = e^{-E(\uparrow)/k_B T} + e^{-E(\downarrow)/k_B T}$$

$$q = e^{-\epsilon/2k_B T} + e^{+\epsilon/2k_B T}$$

Example: Spin System

$$q = e^{-\epsilon/2k_B T} + e^{+\epsilon/2k_B T}$$

- Internal energy $U = -\partial(\ln q)/\partial\beta$

$$U = -\frac{1}{q} \frac{\partial q}{\partial \beta} = -\frac{1}{q} \frac{\partial}{\partial \beta} (e^{-\beta\epsilon/2} + e^{\beta\epsilon/2})$$

$$U = -\frac{1}{q} \left(-\frac{\epsilon}{2} e^{-\beta\epsilon/2} + \frac{\epsilon}{2} e^{\beta\epsilon/2} \right)$$

$$U = -\frac{\epsilon}{2} \frac{e^{\beta\epsilon/2} - e^{-\beta\epsilon/2}}{e^{\beta\epsilon/2} + e^{-\beta\epsilon/2}} = -\frac{\epsilon}{2} \frac{e^{\epsilon/2k_B T} - e^{-\epsilon/2k_B T}}{e^{\epsilon/2k_B T} + e^{-\epsilon/2k_B T}}$$

Example: Spin System

$$q = e^{-\epsilon/2k_B T} + e^{+\epsilon/2k_B T}$$

- Helmholtz energy $A = -k_B T \ln q$

$$A = -k_B T \ln(e^{-\epsilon/2k_B T} + e^{\epsilon/2k_B T})$$

- Entropy $S = (U - A)/T = -(\partial A/\partial T)_V$

$$S = -\frac{\epsilon}{2T} \frac{e^{\epsilon/2k_B T} - e^{-\epsilon/2k_B T}}{e^{\epsilon/2k_B T} + e^{-\epsilon/2k_B T}} + k_B \ln(e^{-\epsilon/2k_B T} + e^{\epsilon/2k_B T})$$

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Classical Statistical Mechanics

What is a **microstate** of a classical particle?



Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$

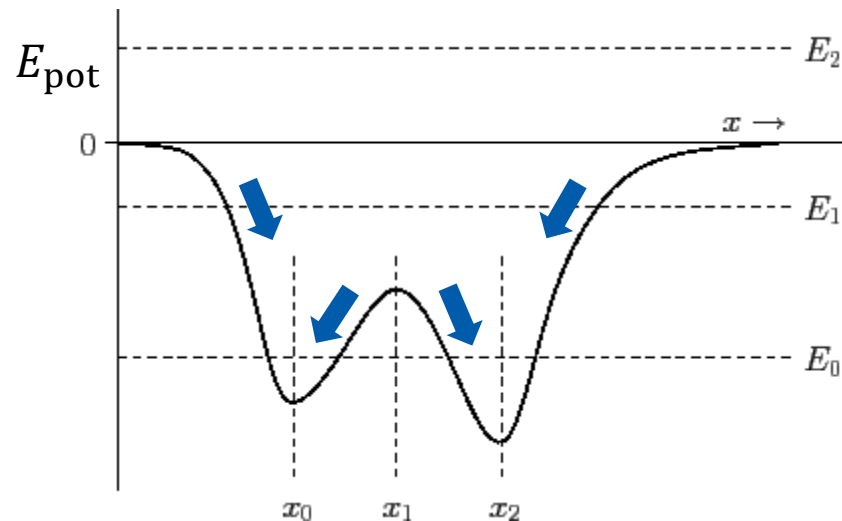
With the velocity, we can define:

- Speed $v = |\vec{v}| = (v_x^2 + v_y^2 + v_z^2)^{1/2}$
- Kinetic energy $E_k = mv^2/2 = m(v_x^2 + v_y^2 + v_z^2)/2$

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Force $\vec{F}$: negative slope of potential energy ($-\nabla E_{\text{pot}}$)



(Image: <http://farside.ph.utexas.edu/teaching/336k/Newtonhtml/node16.html>)

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Acceleration $\vec{a}$: second time-derivative of position

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

$$-\nabla E_{\text{pot}} = m \frac{d^2\vec{r}}{dt^2}$$

↑
given

Second-Order ODE

$$\frac{d^2 \vec{r}}{dt^2} = - \frac{\nabla E_{\text{pot}}}{m}$$

Two conditions are required to obtain the full dynamics:

(initial) position and (initial) velocity (= momentum)

Phase Space

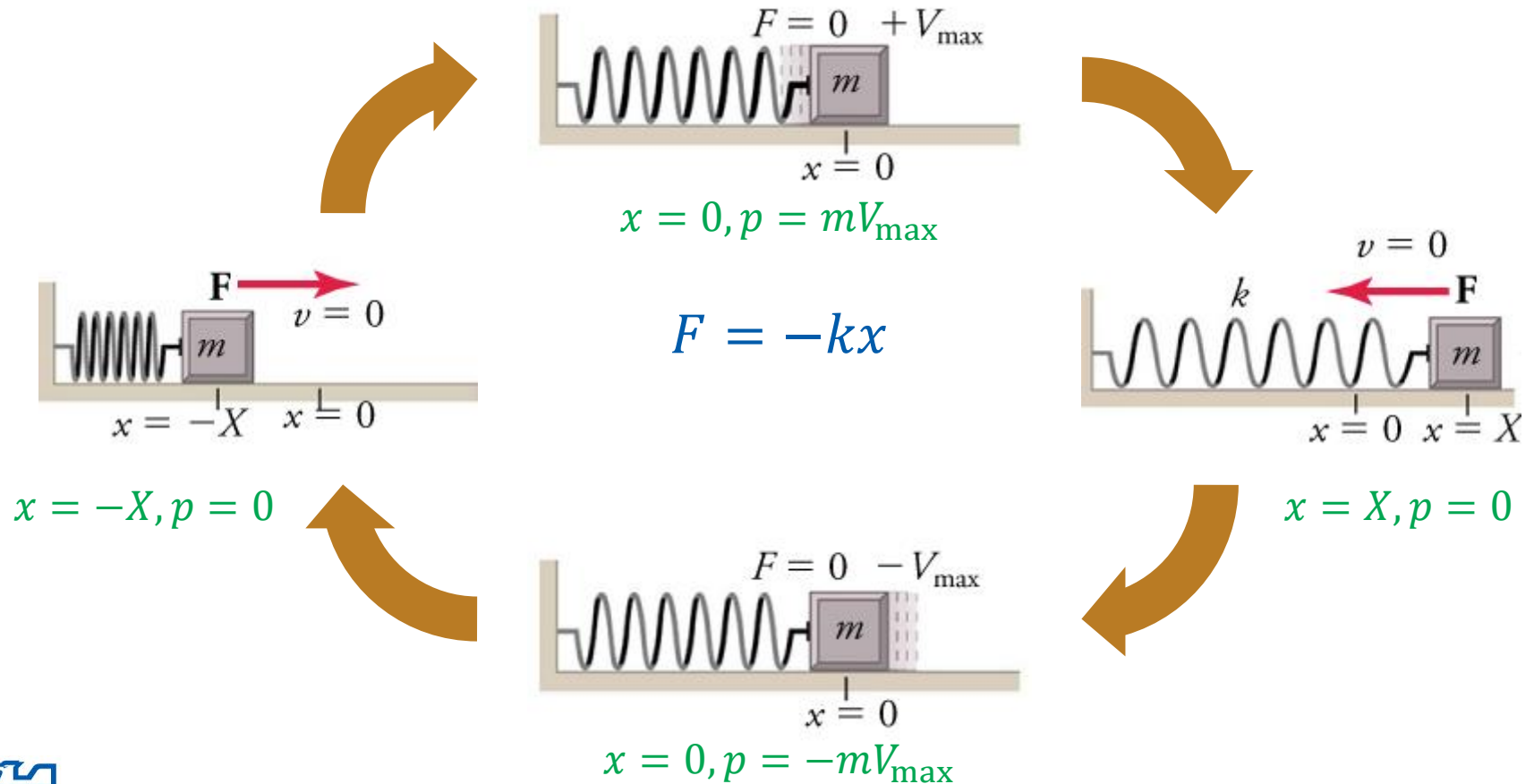
Space of **all possible microstates** of a system

- Classical mechanics:

Space of all possible values of
position and **momentum** variables

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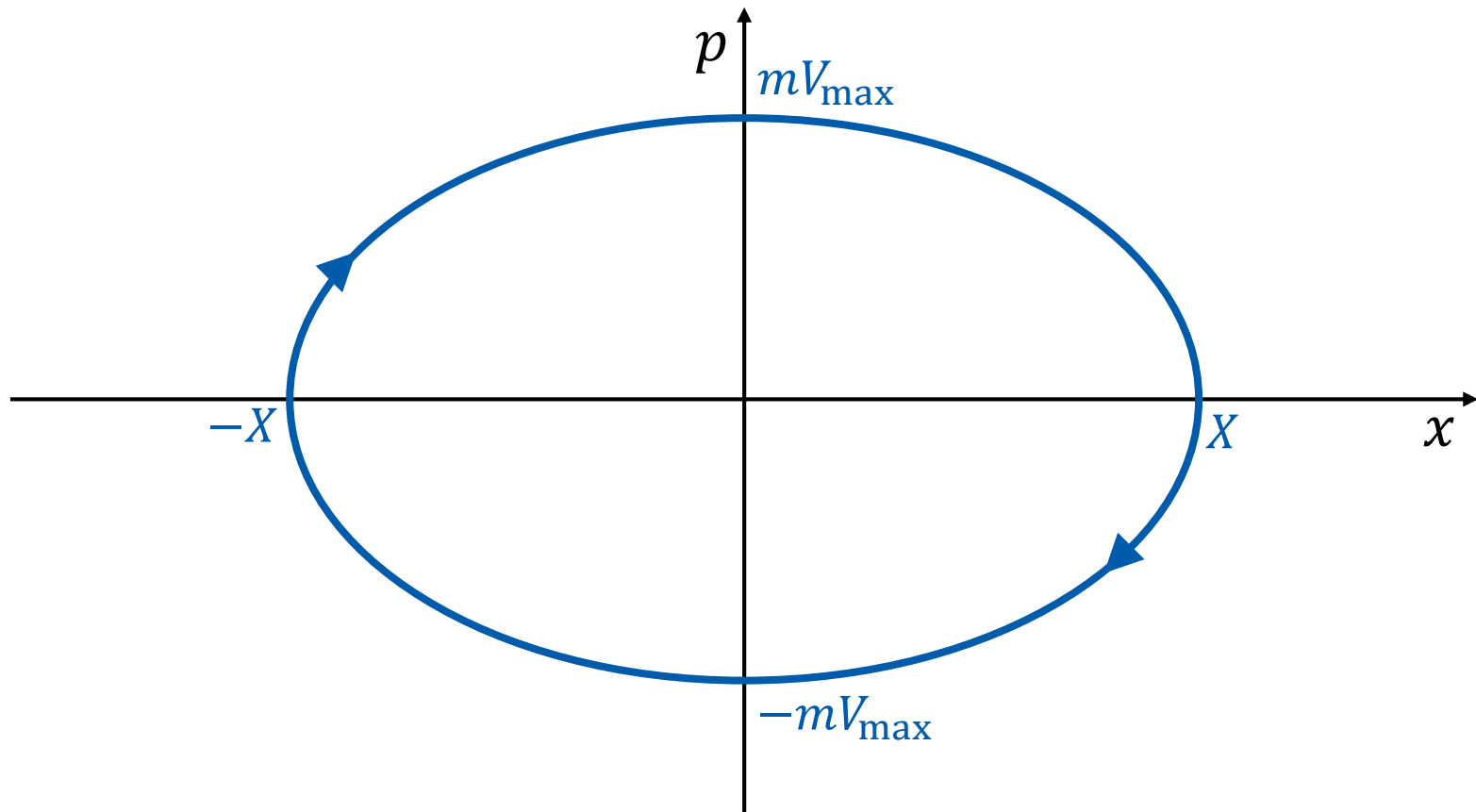
1-Dimensional Spring



(Image: <https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion>)

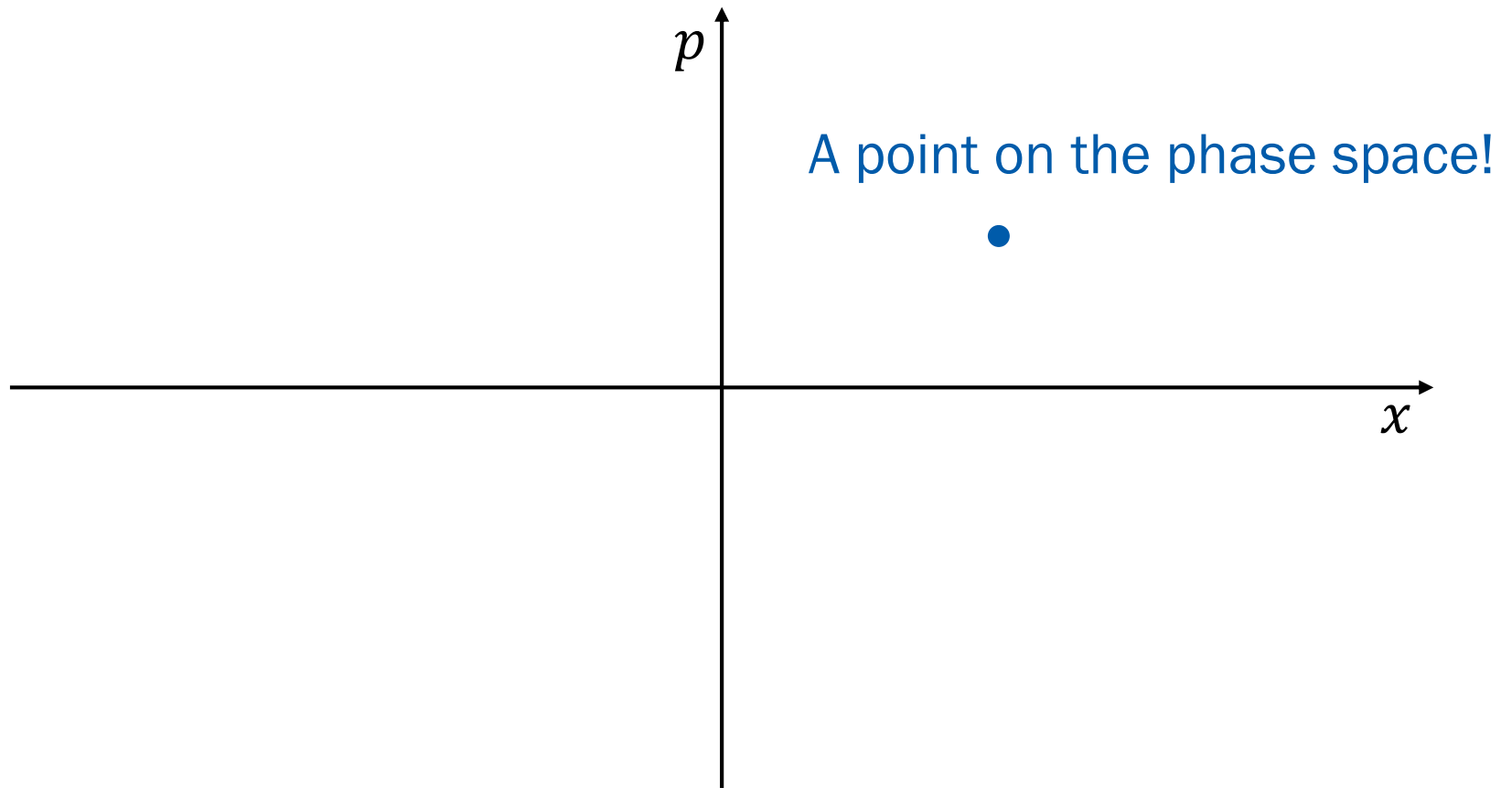
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Phase Space Representation



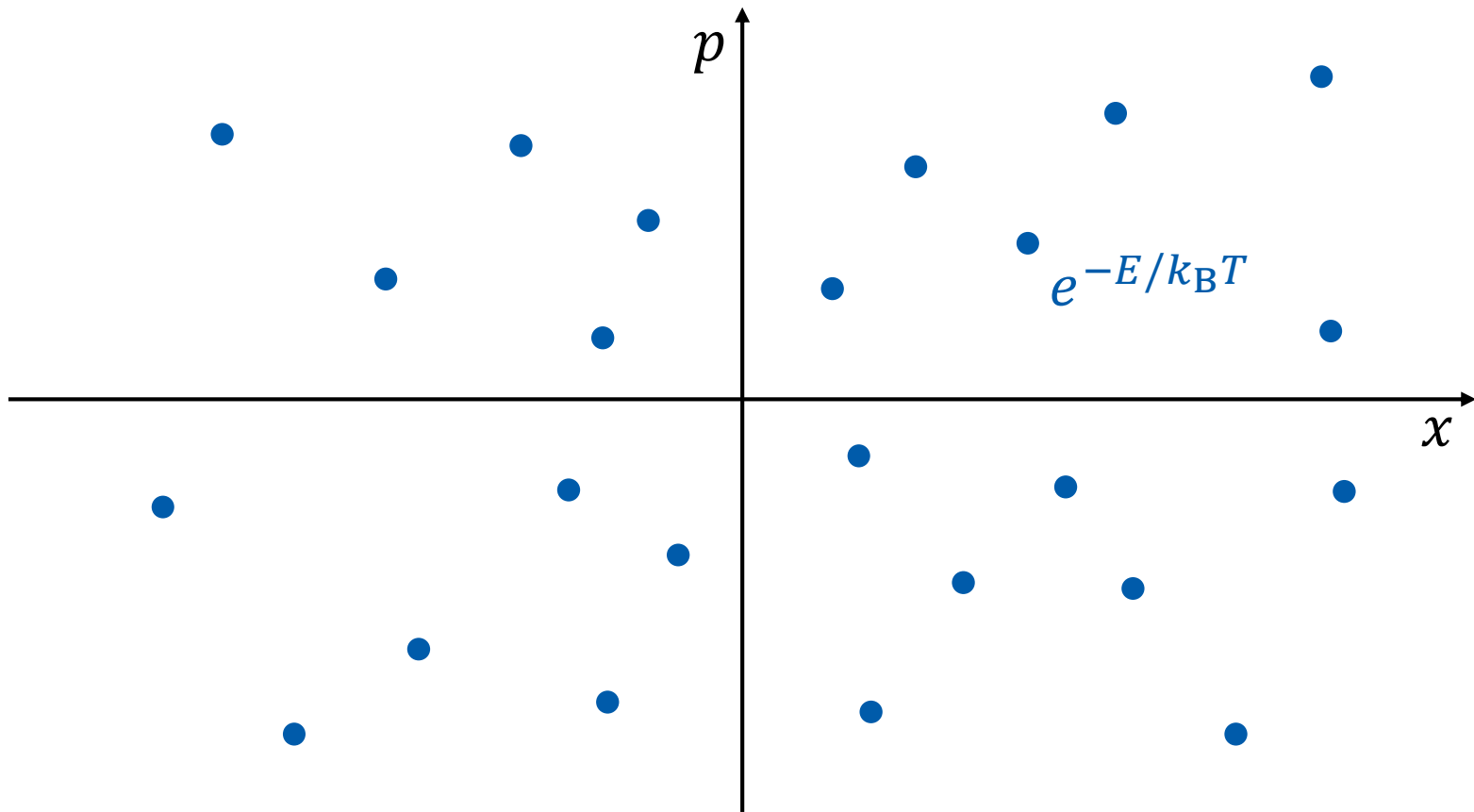
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Microstate of Classical Particle



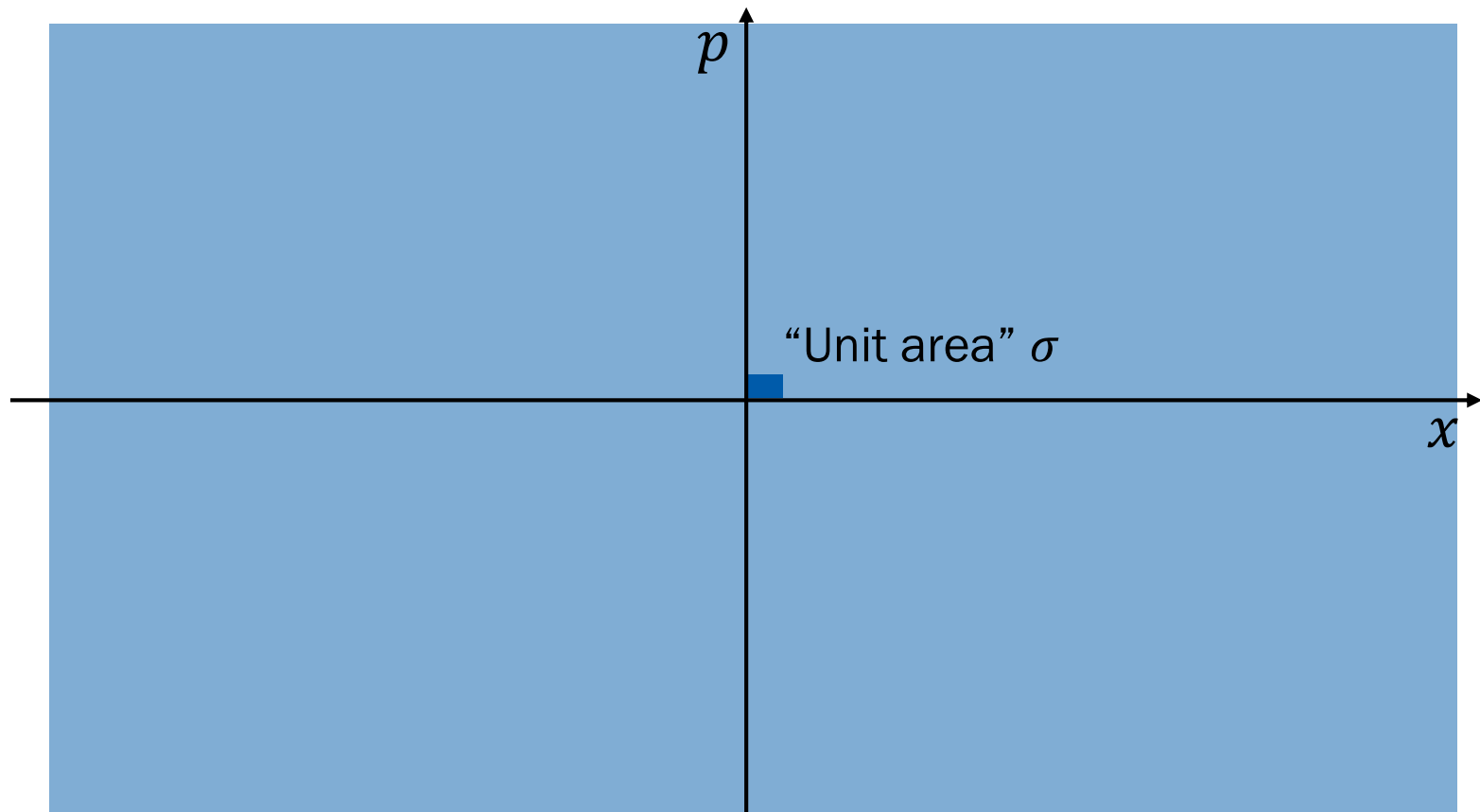
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Partition Function



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Partition Function



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Partition Function

$$q = \sum_s e^{-E(s)/k_B T} \rightarrow q = \frac{1}{\sigma} \int \int e^{-E(x,p)/k_B T} dx dp$$

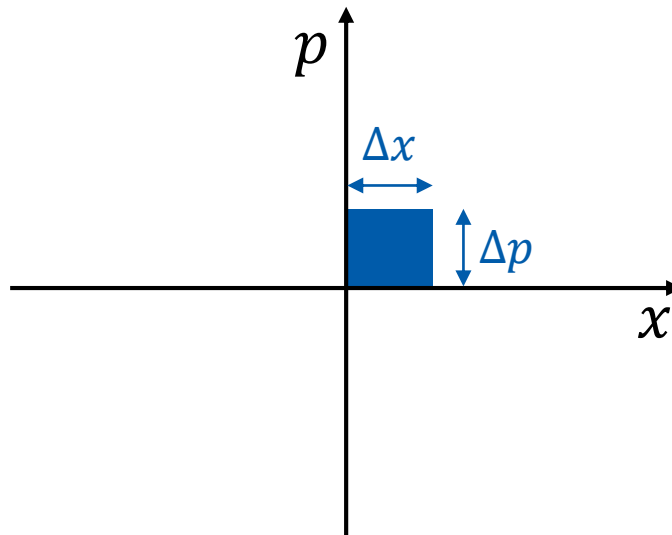
for a 1-dimensional single particle

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What is σ ?

From Heisenberg's uncertainty principle,

$$\Delta x \Delta p \geq h \text{ (Planck's constant)}$$



Take $\sigma = h!$

Partition Function

$$q = \frac{1}{h} \int \int e^{-E(x,p)/k_B T} dx dp$$

- Example: 1-dimensional spring

$$E(x, p) = \frac{1}{2} k x^2 + \frac{1}{2} m v^2 = \frac{1}{2} k x^2 + \frac{p^2}{2m}$$

$$q = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{k_B T} \left(\frac{1}{2} k x^2 + \frac{p^2}{2m} \right)} dx dp$$

1-Dimensional Spring

$$q = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{k_B T} \left(\frac{1}{2} k x^2 + \frac{p^2}{2m} \right)} dx dp$$

$$q = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_B T} - \frac{p^2}{2mk_B T}} dx dp$$

$$q = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_B T}} \times e^{-\frac{p^2}{2mk_B T}} dx dp$$

$$q = \frac{1}{h} \left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_B T}} dx \right] \left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_B T}} dp \right]$$

Gaussian Integrals

$$q = \frac{1}{h} \left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_B T}} dx \right] \left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_B T}} dp \right]$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha},$

$$q = \frac{1}{h} \sqrt{\frac{2\pi k_B T}{k}} \sqrt{2m\pi k_B T} = \frac{2\pi k_B T}{h} \sqrt{\frac{m}{k}}$$

Into Higher Dimensions

- Single 1-dimensional particle

$$q = \frac{1}{h} \int \int e^{-E(x,p)/k_B T} dx dp$$

- Single 3-dimensional particle

$$q = \frac{1}{h^3} \iiint \iiint e^{-E(x,y,z,p_x,p_y,p_z)/k_B T} dx dy dz dp_x dp_y dp_z$$

$$\rightarrow q = \frac{1}{h^3} \int \int e^{-E(\vec{r},\vec{p})/k_B T} d\vec{r} d\vec{p}$$

Higher and Higher

- Single 3-dimensional particle

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

- N 3-dimensional (independent) particles

$$Q = q_1 \times q_2 \times q_3 \times q_4 \times \cdots \times q_N$$

$$Q = \frac{1}{h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N$$

Indistinguishability

- N 3-dimensional distinguishable particles:

$$Q = \frac{1}{h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N$$

- N 3-dimensional indistinguishable particles:

$$Q = \frac{1}{N! h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N$$

Calculation of Q

For indistinguishable particles,

$$Q = \frac{1}{N! h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N = \frac{1}{N!} q^N$$

where

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

Perfect Gas

- No interactions (no potential energy)
- Total energy = kinetic energy ($p^2/2m$)
- Bound in a box of volume V
 - Range of $x = (0, a)$
 - Range of $y = (0, b)$
 - Range of $z = (0, c)$
 - $abc = V$
- Range of momentum = $(-\infty, \infty)$

q of Perfect Gas

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

Since $E(\vec{r}, \vec{p}) = p^2/2m$,

$$q = \frac{1}{h^3} \int \int e^{-p^2/2mk_B T} d\vec{r} d\vec{p}$$

$$q = \frac{1}{h^3} \left[\int 1 d\vec{r} \right] \left[\int e^{-p^2/2mk_B T} d\vec{p} \right]$$

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Position Integral

$$q = \frac{1}{h^3} \left[\int 1 \, d\vec{r} \right] \left[\int e^{-p^2/2mk_{\text{B}}T} \, d\vec{p} \right]$$

$$\int 1 \, d\vec{r} = \int_0^c \int_0^b \int_0^a 1 \, dx \, dy \, dz = abc = V$$

Momentum Integral

$$q = \frac{V}{h^3} \left[\int e^{-p^2/2mk_B T} d\vec{p} \right]$$

$$\int e^{-p^2/2mk_B T} d\vec{p} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-p^2/2mk_B T} dp_x dp_y dp_z$$

Since $e^{-p^2/2mk_B T} = e^{-p_x^2/2mk_B T} e^{-p_y^2/2mk_B T} e^{-p_z^2/2mk_B T}$,

$$\left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_B T} dp_x \right] \left[\int_{-\infty}^{\infty} e^{-p_y^2/2mk_B T} dp_y \right] \left[\int_{-\infty}^{\infty} e^{-p_z^2/2mk_B T} dp_z \right]$$

Gaussian Integral Again

$$q = \frac{V}{h^3} \left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_B T} dp_x \right]^3$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha},$

$$q = \frac{V}{h^3} \left[\sqrt{2\pi mk_B T} \right]^3 = V \left(\frac{2\pi mk_B T}{h^2} \right)^{3/2}$$

Q of Perfect Gas

$$q = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2}$$

For indistinguishable gas particles,

$$Q = \frac{1}{N!} q^N = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Helmholtz Energy

$$Q = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Now, we can calculate the Helmholtz energy:

$$A = -k_B T \ln Q = -k_B T \ln \left\{ \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} \right\}$$

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

Internal Energy

$$Q = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Also the internal energy:

$$U = -\frac{\partial(\ln Q)}{\partial \beta} = -\frac{\partial}{\partial \beta} \ln \left\{ \frac{V^N}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} \right\}$$

$$U = \frac{\partial}{\partial \beta} \ln \beta^{3N/2} + \frac{\partial}{\partial \beta} (\text{terms independent to } \beta) = \frac{3}{2} N k_B T$$

Pressure

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

Using $p = -(\partial A / \partial V)_T$,

$$p = Nk_B T \frac{\partial(\ln V)}{\partial V} + \frac{\partial}{\partial V} (\text{terms independent to } V)$$

$$p = \frac{Nk_B T}{V}$$

One can obtain **entropy** similarly

Real Gas

Particles are no more independent to each other

$$Q \neq q^N / N!$$

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

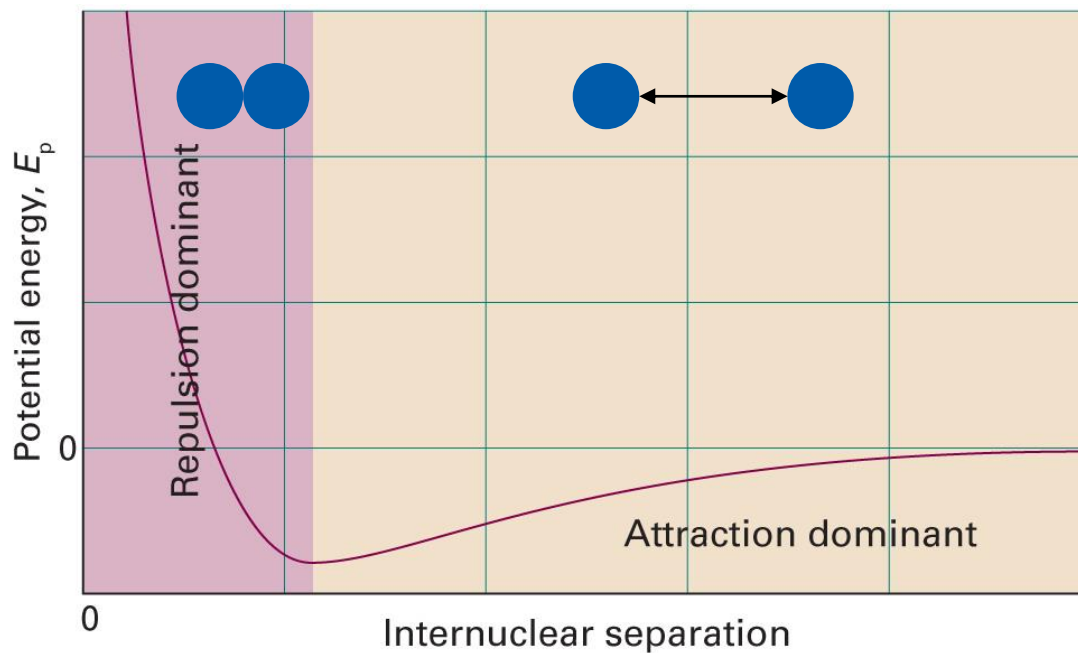
where

$$E_{\text{tot}} = \sum_i \frac{p_i^2}{2m} + U(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$$

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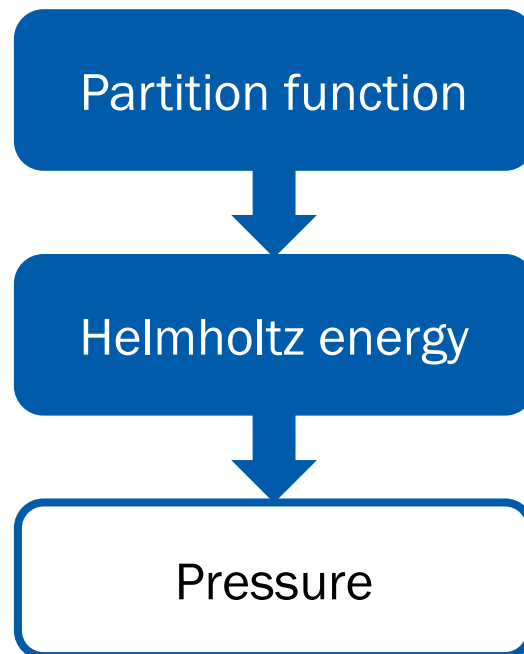
Potential Energy

Pairwise interactions (attraction + repulsion)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 1C.1)

Pressure of Real Gas



$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

Equipartition Theorem

For a sample at thermal equilibrium,
the average value of each **quadratic contribution**
to the internal energy is $(1/2)k_B T$.

(Quadratic contribution: e.g., $E \propto p_x^2$ or $E \propto x^2$)

Can we prove this theorem?

Setting up the Stage

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_{\text{B}}T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Assume that

$$E_{\text{tot}} = \sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2$$
$$(A_i, B_j > 0)$$

and that the ranges of both positions and momenta are $(-\infty, \infty)$

Summation to Product

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-\left(\sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2\right)/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

$$= \frac{1}{N! h^{3N}} \int \cdots \int \prod_{i=1}^{f_r} e^{-A_i r_i^2/k_B T} \prod_{j=1}^{f_p} e^{-B_j p_j^2/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Thus,

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2/k_B T} dr_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_B T} dp_j$$

Gaussian Integrals

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2 / k_B T} dr_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2 / k_B T} dp_j$$

Using

$$\int_{-\infty}^{\infty} e^{-A_i r_i^2 / k_B T} dr_i = \sqrt{\frac{\pi k_B T}{A_i}}, \quad \int_{-\infty}^{\infty} e^{-B_j p_j^2 / k_B T} dp_j = \sqrt{\frac{\pi k_B T}{B_j}},$$

we obtain

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_B T}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_B T}{B_j} \right)^{1/2}$$

In Terms of β

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_B T}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_B T}{B_j} \right)^{1/2}$$

We can write the partition function as follows:

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N! h^{3N}} \left(\frac{1}{\beta} \right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j} \right)^{1/2}$$

Internal Energy

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N! h^{3N}} \left(\frac{1}{\beta}\right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j}\right)^{1/2}$$

The internal energy can be calculated from

$$U = - \left(\frac{\partial \ln Q}{\partial \beta} \right)$$

Now,

$$\ln Q = - \frac{f_r + f_p}{2} \ln \beta + (\text{terms independent to } \beta)$$

Internal Energy

$$\ln Q = -\frac{f_r + f_p}{2} \ln \beta + (\text{terms independent to } \beta)$$

$$U = -\left(\frac{\partial \ln Q}{\partial \beta}\right) = \frac{f_r + f_p}{2} \frac{\partial \ln \beta}{\partial \beta} = \frac{f_r + f_p}{2} \frac{1}{\beta} = \frac{f_r + f_p}{2} k_B T$$

Therefore,

$$U = (f_r + f_p) \frac{k_B T}{2}$$

“The average value of each **quadratic contribution**
to the internal energy is $(1/2)k_B T$.”

Summary

